

Assignment Sheet

Task 1 – HDD Address Modes

(4 Points)

In your own words, explain the difference between CHS and LBA. One of those concepts was used in early disks, why? What changed?

Your answer:

Task 2 – HDD Access Time

(8 Points)

An HDD has the following specifications: It takes $15ms$ to move the r/w head to the correct cylinder, and the total access time is $23ms$. What is this disc's rpm? (Meaning: How many full turns the disc's platters make in one minute.)

rpm=

Your calculations:

Task 3 – Uses of external storage

(8 Points)

From 2016 to 2024 AWS provided a service called Snowmobile to large scale business partners who wanted to ship large amounts of data to their cloud. Instead of sending them via the Internet, the customers could book a truck equipped with a 100 Petabyte data storage, that would download their data and drive it to the AWS data centre.

Based on the data below, calculate the time savings of sending 1 PB of data from New York City to Los Angeles per snowmobile (download data at customer, drive, upload data at AWS) versus the Internet. Ignore all other factors like cost, breaks and delay. Give your result in days.

Internet:

Upload speed: $1GB/s$

Snowmobile:

Capacity: $100PB$

Upload/Download speed (Site - Snowmobile): $1TB/s$

Distance NYC to Los Angeles: 2700 Miles

Snowmobile average speed: 65 Miles per Hour

Saved time:

You calculations:

This assignment is due on 23.01.2026 at 10 am.